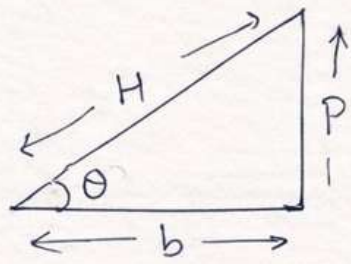
# Maths for AI

## Linear Algebra

i) $\sin\theta = \frac{P}{H}$

ii) $\cos\theta = \frac{B}{H}$

iii) $\tan\theta = \frac{P}{B}$

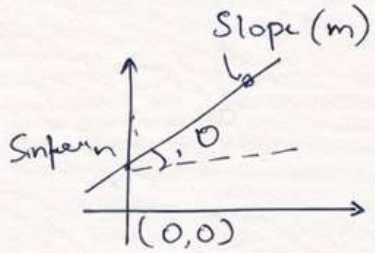



## Straight lines

* eq$^n$ for straight line: $\boxed{y = mx + c}$

where : m = slope

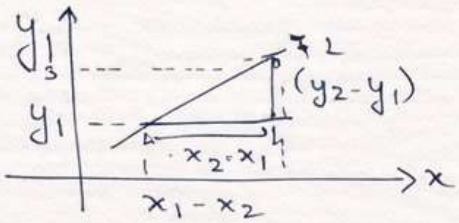
i) c = i : intercept




Eg $P_1(x_1, y_1) \rightarrow (2, 2, y_1)$

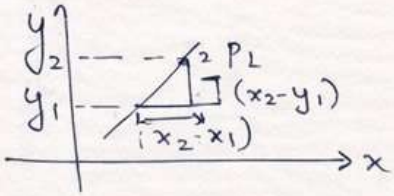
$P_2(x_2, y_2) \rightarrow (5 = 10)$

$$\tan\theta = \frac{P}{B} = \boxed{\frac{y_2 - y_1}{x_2 - x_1} = m}$$




## Distance between two points

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

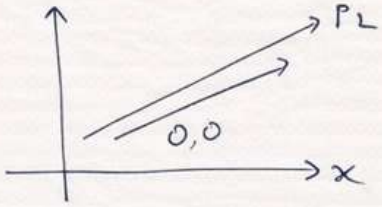



## Parallel & Perpendicular lines

### Parallel lines

L1: $y = m_1 x + c_1$

L2: $y = m_2 x + c_2$ $(m_1 = m_2$

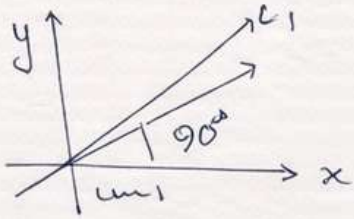



### Perpendicular lines

L1: $y = m_1 x + c_1$

L2: $y = m_2 x + c_2$

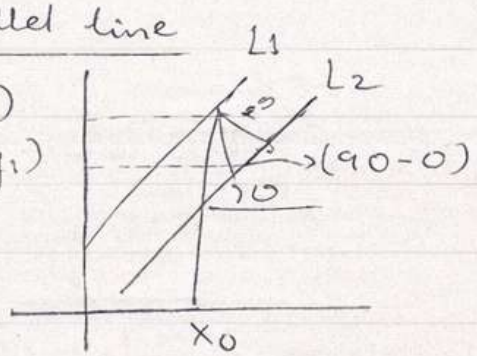
$\boxed{m_1 m_2 = -1}$



## Distance btw 2 parallel line

$L1 = y = mx + c_1 \ (x_0, y_1)$

$L2 = y = mx + c_2 \ (x_0 - y_1)$




$$\boxed{\cos\theta = \frac{d}{|C_2 - C_1|}}$$

$$\sin\theta(0-\theta) = \frac{P}{T_1} = \frac{d}{|C_2 - C_1|} = L_0 + \theta$$

$$T_2 = \boxed{d = \frac{?}{\sqrt{1+m^2}} = \cos\theta}$$

Rmk

$$\boxed{d = \frac{|C_2 - C_1|}{\sqrt{1+m^2}}}$$

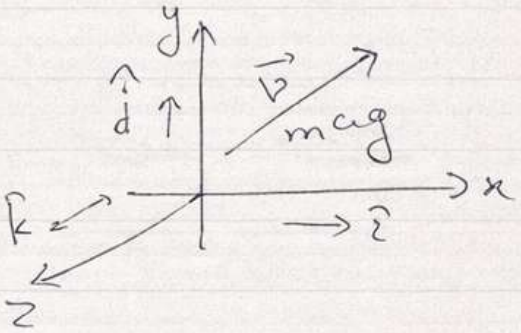
## Vecton ($\vec{v}$)

(1) It is a quantity that has both both magnitude (size/length), & direction

$$\boxed{\vec{v} = \vec{v_1} + v_2\hat{j} + v_3\hat{k}}$$

$$|\vec{v}| = \sqrt{v_1^2 + v_2^2 + v_3^2}$$

distance from origin(0,0,0)



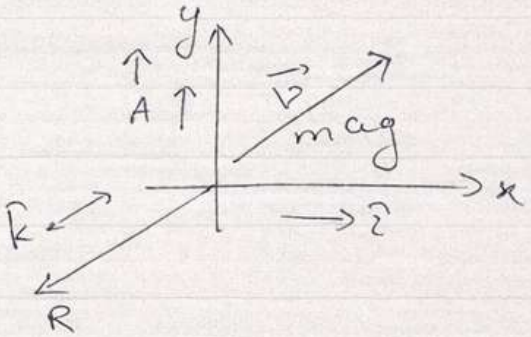

## Vector operation

i) vector addition

$A = (2,1,3)$

$B = (4,5,2)$

$$\boxed{R = 6\hat{i} + 6\hat{j} + 5\hat{k}}$$

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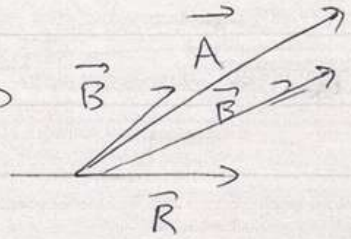



$$\boxed{R = 6\hat{i} + 6\hat{j} + 5\hat{k}}$$

vi) Scalar multiplication

$A = (2,1,3) = (6,3,9)$ $\quad \vec{6} \times 3$

$$\boxed{6\hat{i} + 3\hat{j} + 9\hat{k} = (6,3,9)}$$

vii) Dot product

$\vec{A} \cdot \vec{B} = a_1 b_1 + a_2 + b_2$

$$\boxed{\vec{A} \cdot \vec{B} = |A||B| \cos\theta}$$

(orthogonal)

$A \cdot B = 0$ · vector

$\cos 90° = 0$

viii) Projections)

C.) Cross product

$$\boxed{\vec{A} \cdot \vec{B} = |a_1 b_1 + a_2 + a_3 3}$$

(orthogonal)

↳ produces a new vector ⊥ both -⊥ vector to both original vectors – into the plane

$$A \times \begin{vmatrix} \vec{i} & \vec{k} \\ a_1 & a_1 \\ b_1 & b_1 \end{vmatrix} \begin{matrix} a_2 & a_3 \\ b_2 & b_3 \end{matrix}$$

Matrix (Plural = Matrices)

(i) If is a rectangular arrey ∫ numbers, symbols, or expressions, arranged in rows and columns:

$$\begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$$

Matrix operations

i) matrix addition / subtraction

ii) scalar multiplication (KA) ⟶

iii) matrix multiplication (AB)

iv) Transpose (AT)

v) Dot product (of row matrix & column matrix

note;

$(M \times N)$, $(N \times P)$

(matrices) $A \cdot B = C$

size, $(C = AB) \Rightarrow (M \times P)$

## Determinants ($|A|$)

is a unique number calculated from a square matrix:

$$|A| = \det(A) = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \quad a\begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

$$3\times3 \quad = \begin{vmatrix} \vec{a} & \vec{a} = \vec{a}_2 \vec{a} \\ a & f \quad g \quad f \\ g & c_3 \quad d \quad k \end{vmatrix} \quad \left\{ \begin{matrix} \text{even} = + \\ \text{odd} = - \end{matrix} \right\}$$

calculate minors

$m_1 = ei = fh$

$m_2 = di - fg$ $\qquad |a_1| = |a_1 b_1 + - m_2 m m_3$

$m_3 = dh - eg$ $\qquad$ [~~Cofactor(ij) = jk jc~~]

$$\left[ \begin{vmatrix} \vec{e} & \vec{f} \\ h & i \end{vmatrix} \rightarrow \begin{bmatrix} \vec{i} \\ i \end{bmatrix} \right] \rightarrow \int \text{2D plane)}$$

$$a_{12}x_1 + b_{12}x + c_{13}x_3 = d_1$$
$$a_{21}x_1 + b_{22}x + c_2x_3 = d_2$$
$$a_{31}x_1 + b_{32}x + c_3x_3 = d_3$$

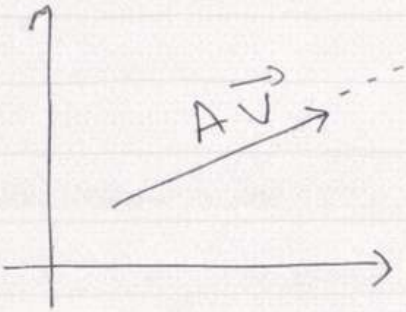
$$\left.\begin{matrix} a_{11}x_1 + b_{12}x_2 + c_{13}x_3 = d_1 \\ a_{21}x_1 + b_{22}x_2 + c_2x_3 = d_2 \\ a_{31}x_1 + b_{32}x + c_3x_3 = d_3 \end{matrix}\right\}$$

$$\begin{vmatrix} a_{11} & b_{12} & b_{13} \\ a_{21} & b_{32} & b_{33} \\ a_{31} & b_{32} & b_{33} \end{vmatrix} \neq 0 \quad \text{(unique Aol}^n\text{)}$$

# Eigen vectors

If $\lambda$ is a n-zero vector that when the matrix acts on it does not change directions – it only gets scaled.

$$\begin{vmatrix} a_{11} & b_{12} & b_{13} \\ a_{21} & b_{32} & b_{23} \\ a_{31} & b_{32} & b_{33} \end{vmatrix} \neq 0 \quad \text{(unique Aol}^n\text{)}$$




(scaled upward or downward)

(not change direction)

$$\boxed{AV = \lambda V}$$

$\lambda$ → eigen value (scalar)

$V$ → eigen vector (vector)

$A$ → square

$$\boxed{(A - \lambda I)V = 0}$$

$$|A - \lambda I| = 0$$

$\lambda$ → eigen value

$$|A - \lambda I| = 0$$

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